

Climate changes and trends in phenology of fruit trees and field crops in Germany, 1961–2000

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Received 21 February 2003; received in revised form 17 July 2003; accepted 25 July 2003

Abstract

Distinct changes in air temperature since the end of the 1980s have led to clear responses in plant phenology in many parts of the world. In Germany phenological phases of the natural vegetation as well as of fruit trees and field crops have advanced clearly in the last decade of the 20th century. The strongest shift in plant development occurred for the very early spring phases. The late spring phases and summer phases reacted also to the increased temperatures, but they usually show lower trends. Until now the changes in plant development are still moderate, so that no strong impacts on yield formation processes were observed. But further climate changes will probably increase the effect on plants, so that in the future stronger impacts on crop yields are likely.

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Keywords: Climate change in Germany; Phenology; Fruit trees; Field crops

1. Introduction

The distinct increase in air temperature in many parts of the northern hemisphere since the end of the 1980s and the demand for indicators of climate change impacts has caused a growing interest in phenological data. From this time onwards, phenological data were more and more accepted by climate researchers.

Since then many papers report on trends in the timing of spring events in mid- and high latitudes (Braslavska and Kamensky, 1999; Schwarz and Reiter, 2000; Defila and Clot, 2001; Müller, 2002)

and on relationships between air temperature changes and variations in the timing of phenological stages (Bradley et al., 1999; Beaubien and Freeland, 2000; Sparks et al., 2000; Schmerbach, 2000; Chmielewski and Rötzer, 2002). Most papers focus on changes in the natural vegetation, but only a few studies deal with trends of agricultural and horticultural varieties (Schelling, 2000). However, changes in the timing of phenophases of fruit trees or field crops could be of great economical importance, because they could have direct impacts on yield formation processes and so on the final crop yield.

In mid- and high latitudes, with a vegetation-rest in winter (dormancy) and an active growing period in summer, plant phenology is mainly driven by temperature. After the release of dormancy the development of plants strongly depends on air temperature. With

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increasing temperature the biochemical reactions are accelerated up to a threshold where enzyme systems are destroyed and cells die.

Thus, an indirect effect of increasing air temperatures is the prolongation of the growing season (Menzel and Fabian, 1999; Chmielewski and Rötzer, 2001) and the modification of the phenological phases of individual plants, i.e. the sequence of the development stages. The extension of the growing season will have mostly positive effects on crop farming in the mid- and high latitudes, since a longer growing season will improve the scope for cultivars selection, catch cropping and crop rotation. Shorter developmental periods for field crops could have rather negative effects on the formation of the individual yield components, as for cereals the crop density, the kernel number per ear, and the kernel weight.

The aim of this study is to show whether fruit trees and field crops show similar trends in plant development as the natural vegetation and if these trends are also related to climate changes.

2. Data

2.1. Phenological data

For this study, mainly the phenological observations of fruit trees (apple, sweet cherry) and field crops (winter rye, sugar beet and maize) from the German Weather Service (DWD) between 1961 and 2000 were used. The phenological network of the DWD consists of about 1500 observers, who do the observations on a voluntary basis (Bruns, 2001).

For apple (*Malus domestica*) and sweet cherry (*Prunus avium*), flowering dates of early maturing varieties were selected. The beginning of flowering was defined as the time in the year when at some places of the plant the first flowers have opened completely (BAHP, 1991).

Winter rye (*Secale cereale*) is one of the crops for which phenological observations in Germany have been made in detail and which are available for the whole country. For sugar beet (*Beta vulgaris*) and maize (*Zea mays*) only data from the main cropping areas in Germany were available. These are for sugar beet data from the federal states Lower Saxony, Bavaria, North-Rhine Westphalia and for maize obser-

vations from Lower Saxony, North-Rhine Westphalia, Baden Württemberg, as well as Bavaria.

To define the average beginning of growing season in Germany, data from the International Phenological Gardens (IPG) in Europe were used. In this study the beginning of growing season (B_G) was defined in the same way as for Europe in Rötzer and Chmielewski (2001). For that study B_G was represented as the average date of leaf unfolding of white birch, wild cherry, mountain ash and alpine currant. The selection of plants will not affect the results which were achieved, because the investigations will show that all phenological phases in spring are well correlated to each other. This time only phenological data from the 22 German IPG-stations in the period of 1961–2000 were considered to calculate this average leaf unfolding index.

2.2. Climatic data

In order to describe the relationships between temperature and plant development, station data of air temperature from the *World Weather Records* (WWR) were used to calculate the average air temperature for Germany. Altogether 16 stations were considered which ranges in elevations between 4 m (Rostock, NE-Germany) and 462 m (Munich) and which are well distributed across Germany. Since air temperature is uniform over relatively large areas, 16 stations should be sufficient to calculate average monthly temperatures for Germany. For example the seasonal air temperature at Frankfurt a.M. is a representative of up to a radius of 500 km around this station (Schönwiese et al., 1998).

3. Results

3.1. Air temperature changes in Germany

For the standard period of 1961–1990, the average annual air temperature in Germany is 8.6 °C. In the period of 1961–2000 the temperature increased continuously (Fig. 1). The linear trend in the data is +0.36 °C per decade ($P < 0.01$). This means that for the whole period (40 years) the observed warming is 1.4 °C. The most interesting feature in this time series is the relatively strong change in air temperature since the end

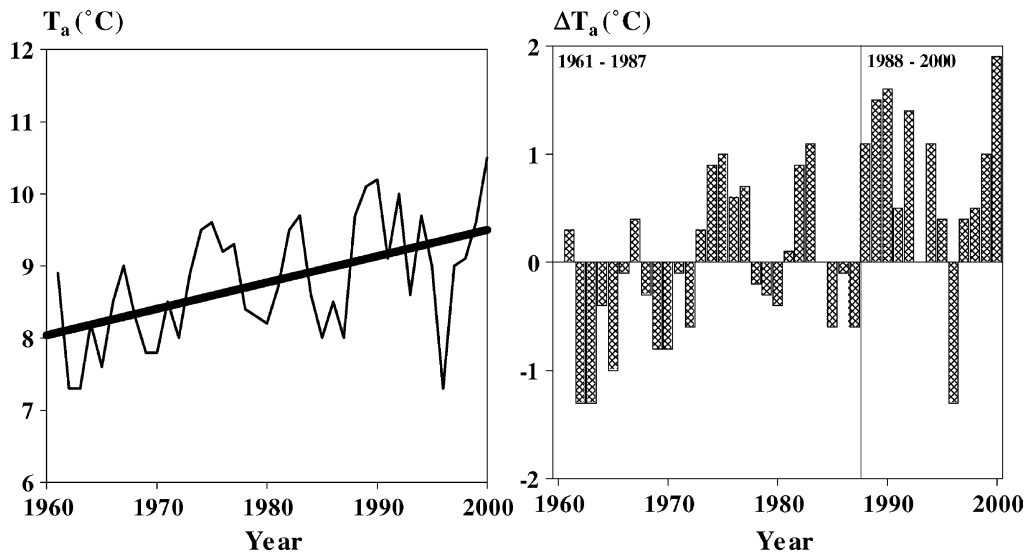


Fig. 1. Average annual air temperature (T_a) for Germany, 1961–2000, left: annual means, right: anomalies compared to the 1961–1990 standard period.

of the 1980s, which corresponds well to similar trends in many parts of the world (Houghton et al., 2001). Since then nearly all years were warmer than average. The only exception in Germany was in 1996, because of a relatively cold winter in 1995–1996.

Many phenological phases, such as the leafing of trees and the flowering of fruit trees are observed in Germany in April and May. April is the time of the year when the beginning of growing season usually starts. The variability of weather from year to year leads to certain fluctuations in the annual timing of phenophases. Previous investigations showed that for the timing of spring phases the average air tempera-

ture from February to April (T_{24}) is a good climatic indicator (Chmielewski and Rötzer, 2001).

The average increase of air temperature in these months ($+0.41^\circ\text{C}$ per decade, $P < 0.05$) is slightly larger than that calculated for the annual average (Table 1). The strongest trend was found for March ($+0.58^\circ\text{C}$ per decade). Rapp and Schönwiese (1996) mentioned the strong trends in air temperature from December to March in Germany, as well.

The anomalies of T_{24} from the standard period of 1961–1990 show the same abrupt change since 1988 as shown before for the annual average. Again, nearly all years since 1988 were warm in this period (Fig. 2).

Table 1

Statistical parameters and trends of average monthly and annual air temperature in Germany, 1961–2000^a

	Month											
	01	02	03	04	05	06	07	08	09	10	11	12
$\bar{x}$	0.4	1.2	4.3	8.0	12.6	15.6	17.4	17.0	13.6	9.2	4.4	1.5
s	2.7	2.6	2.0	1.2	1.4	1.1	1.7	1.4	1.4	1.3	1.5	2.1
Trend ($^\circ\text{C}$ per decade)	+0.59	+0.37	+0.58	+0.28	+0.53	+0.09	+0.48	+0.64	+0.13	+0.10	−0.01	+0.60
P	—	—	0.05	0.10	0.01	—	0.05	0.01	—	—	—	0.05
Year	$\bar{x} = 8.8^\circ\text{C}$, $s = 0.8^\circ\text{C}$, Trend = $+0.36^\circ\text{C}$ per decade ($P < 0.01$)											

$\bar{x}$: mean, s : standard deviation, P : error probability.

^a Average of 16 stations from the World Weather Records.

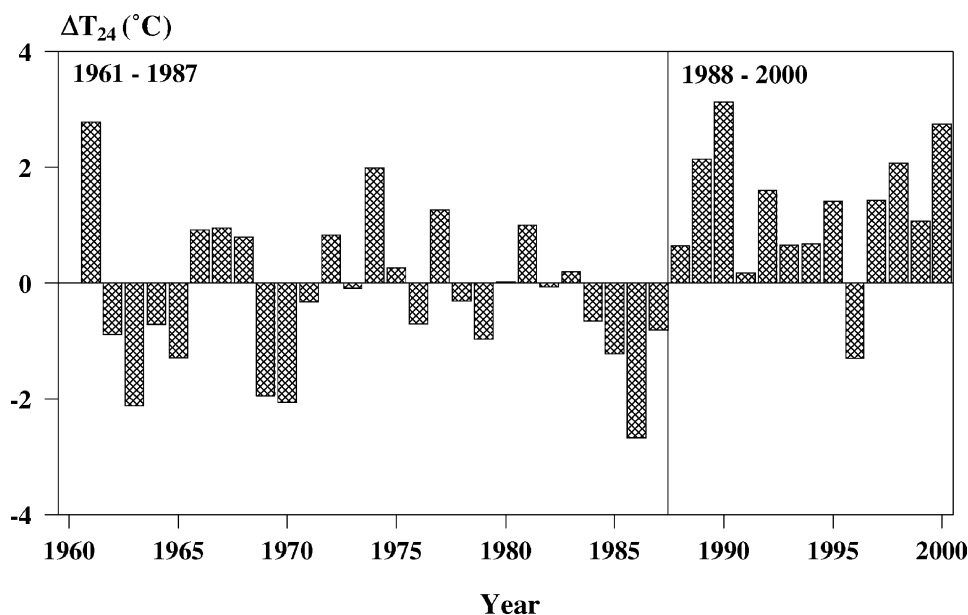


Fig. 2. Average air temperature from February to April for Germany, 1961–2000, anomalies (ΔT_{24}) compared to the 1961–1990 standard period.

3.2. Changes in the timing of phenological events

In this section trends in the beginning of growing season (B_G), in the beginning of blossoming (BBCH¹ 60) of apple (A_{60}) and cherry (C_{60}) trees as well as in the beginning of stem elongation (BBCH 31) for winter rye (R_{31}) were investigated. All these phenophases are observed between the end of April and the beginning of May (Table 2). The standard deviation of these phases is of the same order and the trends for all phases are significant, ranging between 2 and 3 days per decade. This means that the plants react very similar to the environmental conditions in the early spring.

The correlation coefficients (r) between the individual phases are all significant ($P < 0.01$) and range between 0.88 ($r(B_G, R_{31})$) and 0.96 ($r(A_{60}, B_{60})$). A good correlation also exists between the beginning of

growing season and the tree blossom of apple ($r = 0.92$) and sweet cherry ($r = 0.95$) in Germany. This shows that B_G is a good indicator for other phenological events.

Toward the end of the investigated period clear changes in the timing of all phenological events are discernible (Fig. 3) which explain the negative trends in Table 2. These earlier dates seem to correspond to the observed rises in air temperature since the end of the 1980s.

Table 2
Statistical parameters for the timing of different phenophases in Germany, 1961–2000

P	n	x	s	Trend in P (days per decade)
B_G	22	20/4	7.3	−2.3**
R_{31}	637	26/4	6.0	−2.9***
C_{60}	1068	27/4	7.6	−2.0**
A_{60}	908	5/5	7.5	−2.2**

P : phenophase (B_G : beginning of growing season, R_{31} : beginning of stem elongation for winter rye, C_{60} : beginning of cherry tree blossom, A_{60} : beginning of apple tree blossom), n : number of stations used, x : average date, s : standard deviation. Trends are significant with * $P < 0.10$, ** $P < 0.05$, *** $P < 0.01$.

¹ BBCH: Biologische Bundesanstalt, Bundessortenamt, Chemische Industrie (Federal Biological Research Centre for Agriculture and Forestry, Federal Office of Plant Varieties, Chemical Industry). This code is recommended for phenological observations, Strauß et al. (1994).

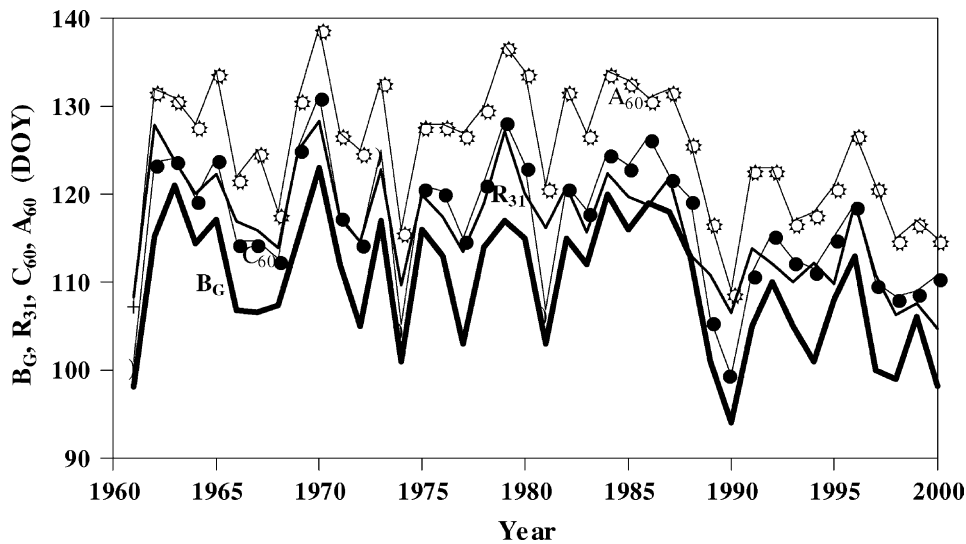


Fig. 3. Average dates of the beginning of growing season (B_G), beginning of stem elongation for winter rye (R_{31}), beginning of cherry tree blossom (C_{60}) and beginning of apple tree blossom (A_{60}) in Germany, 1961–2000. DOY: day of the year.

3.3. Relationships between trends in air temperature and phenophases

As expected, all phenophases are well correlated to the average air temperature from February to April (Table 3). The correlation coefficients range between -0.85 and -0.89 , indicating that higher temperatures after the winter rest of plants accelerate the developmental processes and finally lead to an advanced timing of spring events. The regression coefficients show that an increase in average air temperature between February and April of 1°C leads to an advanced beginning of growing season and blossoming of fruit trees by about 5 days. Only the beginning of shooting

of winter rye shows a smaller temperature response of -3.8 days per 1°C .

Another interesting feature of the advanced timing of spring events is shown in Fig. 4. The changes in plant phenology occurred abruptly since the end of the 1980s, right in the time when the temperatures increased in a similar way (see Fig. 2). The abrupt changes in the timing of phenophases were observed for all species in Table 3. Just as the natural vegetation, fruit trees and field crops show the same behaviour that is shown in Fig. 4.

As an impressive example, the response of the beginning of apple tree blossom to the annual deviations in air temperature is presented in Fig. 5. Years with temperatures above normal in late winter and early spring are clearly related to negative anomalies in the date of blossom. Interesting are the years 1961 and 1990 which are characterised by very early dates in the beginning of apple tree blossom. Both years show comparable deviations in air temperature of about $+3^\circ\text{C}$.

In order to investigate whether the changes in apple tree phenology in the 1990s were observed in any time before since reliable records of air temperature are available, we used the regression equation between air temperature (T_{24}) and the blossoming date (A_{60}) to estimate the beginning of apple tree blossom for

Table 3

Correlation coefficients (r) between the average air temperature from February to April (T_{24}) and the timing of phenophases (P) in Germany, 1961–2000

P	$r(P, T_{24})$	Temperature response ($\Delta P / \Delta T_{24}$) (days/ $^\circ\text{C}$)
B_G	-0.89^{***}	-4.7^{***}
R_{31}	-0.86^{***}	-3.8^{***}
C_{60}	-0.88^{***}	-4.7^{***}
A_{60}	-0.85^{***}	-4.6^{***}

Sensitivity of phenophases to the air temperature changes (ΔT_{24}), significance level: $*** P < 0.01$.

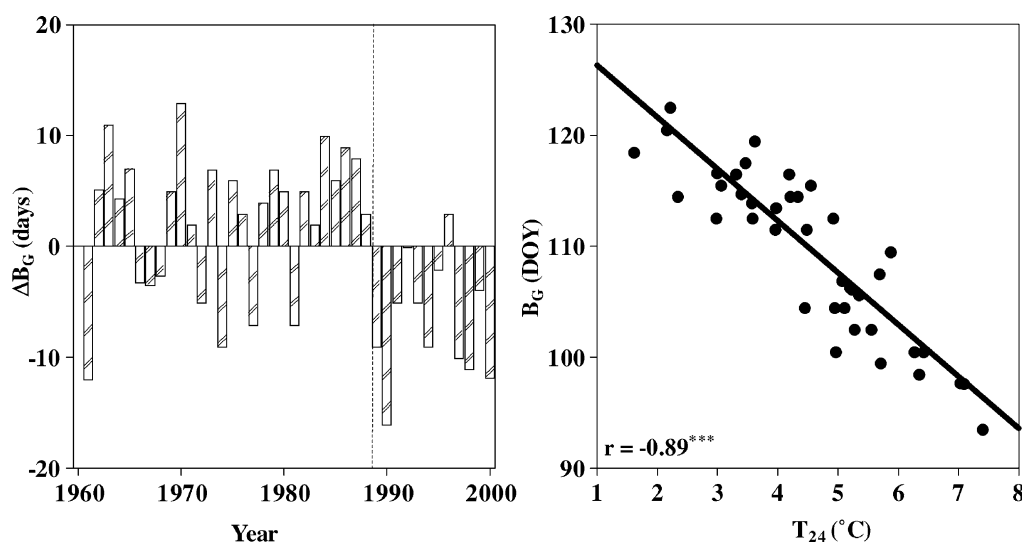


Fig. 4. Left: anomalies in the beginning of growing season (ΔB_G) for Germany (German stations of the International Phenological Gardens) and right: correlation between B_G and the average air temperature from February to April (T_{24}), 1961–2000. The correlation coefficient of $r = -0.89$ is significant with $***P < 0.01$. DOY: day of the year. The vertical line indicates the beginning of distinct changes in B_G .

Germany since 1761 (Fig. 6). Long-term observations of this event are unfortunately not available for Germany, so that we used these proxy-data. The result shows that the recent changes in the timing of cherry

tree blossom seem to be the strongest in the last 240 years. In no decade before as in the 1990s such early dates were observed (average blossom of apple trees in the 1990s: day 121, $s = 5.4$; 1761–1899: day 129,

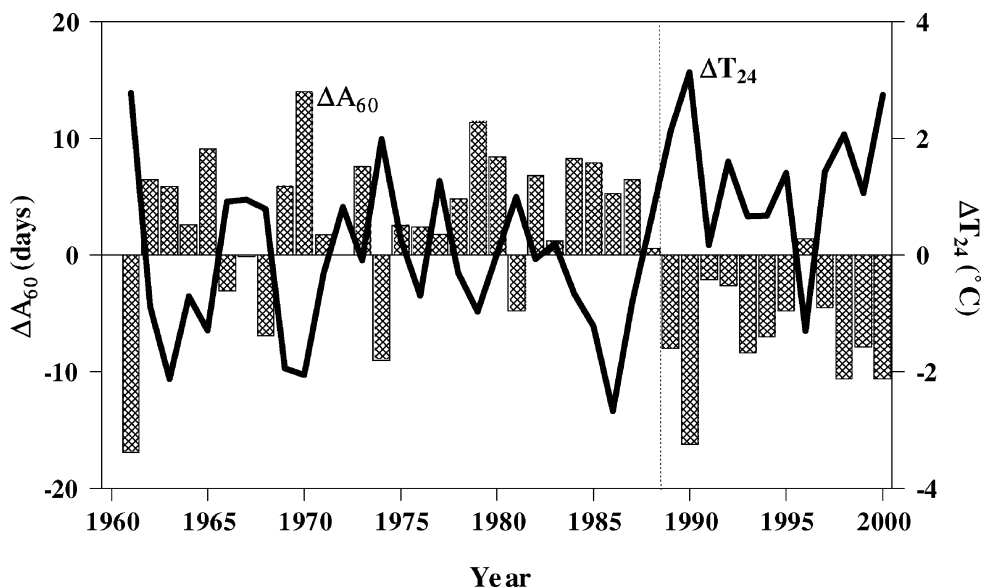


Fig. 5. Anomalies in the beginning of apple tree blossom (ΔA_{60}) and in the average air temperature from February to April (ΔT_{24}) in Germany, 1961–2000. The regression equation between the time series is $A_{60} = 145.9 - 4.56 T_{24}$. The vertical line indicates the beginning of distinct changes in A_{60} .

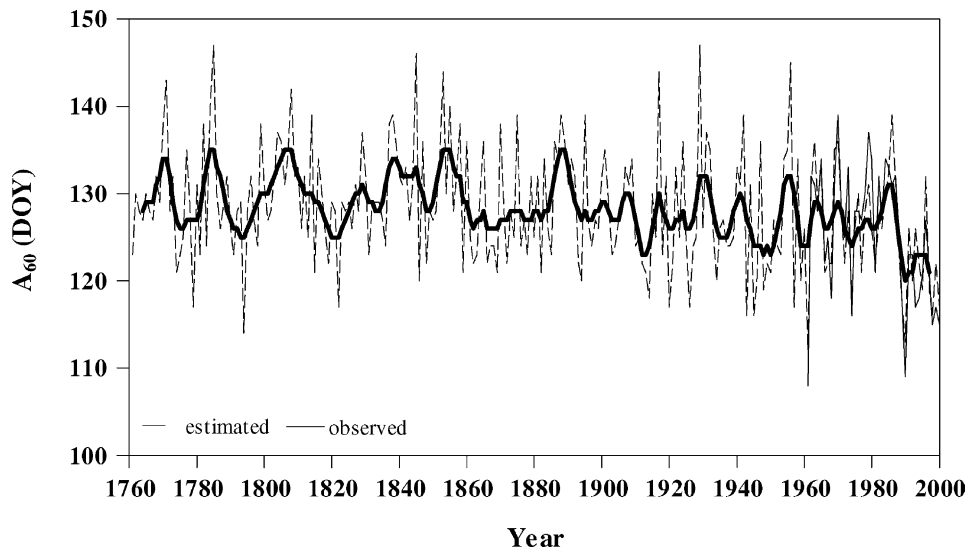


Fig. 6. Estimated (dashed line, 1761–2000) and observed (solid line, 1961–2000) dates of the beginning of apple tree blossom in Germany. The bold line represents an 11 year running mean (Gaussian filter), which shows the long-term variations in this time series. The time series of air temperature was taken from [Rapp \(2000\)](#).

$s = 6.5$). Thus, the timing of this event has advanced more than 1 week.

At the end of this section once again we will focus on the phenology of winter rye in Germany and additionally for selected regions on the phenology of maize and sugar beet. For winter rye very detailed phenological observations during the growing time are available. All phenophases from seeding to harvest show negative trends ([Table 4](#)). The strongest trend occurs for the beginning of stem elongation after the tillering period in the end of April. For the following phases until flowering the negative linear trends decrease slightly and also the significance level reduces. This indicates that the early phases depend most highly on the course of air temperature. With generally increasing temperatures in May and June the mean values now are closer to the optimum for the plant development, so that the plants do not react so sensitive to the variability of air temperature in this time of the year as in the early spring.

Because of shifts of individual phenological phases, the developmental periods of winter rye are slightly shortened, as the period from seeding to emergency (BBCH 00–09), or extended as the shooting period (BBCH 31–51). Such changes in the duration of phenological periods directly influence the yield

formation of winter rye ([Römer, 1988](#); [Petr, 1991](#); [Chmielewski and Köhn, 2000](#)). In spring an early beginning of shooting (BBCH 31) ends the tillering period earlier. A high number of tillers before

Table 4

Statistical parameters of winter rye phenology in Germany, 1961–2000 and calculated trends

Phase (BBCH- code)	x (DOY)	Date	s (days)	Min (DOY)	Max (DOY)	Trend (days/10a)
R_{00}	275.1	2/10	2.72	269	282	−1.4***
R_{09}	288.2	15/10	3.29	281	298	−1.6***
R_{31}	116.4	26/4	6.20	105	128	−2.9***
R_{51}	140.9	21/5	5.00	129	151	−2.0***
R_{65}	157.7	7/6	4.77	146	167	−1.4**
R_{89}	219.1	7/8	6.67	209	232	−1.5*
Duration of phases						
R_{00} – R_{09}	13.1		0.88	12	16	−0.3**
R_{31} – R_{51}	24.5		2.44	21	31	+1.0***
R_{51} – R_{65}	16.8		2.69	11	24	+0.6
R_{65} – R_{89}	61.4		5.03	52	78	−0.1

x: mean, s: standard deviation, min: extreme minimum, max: extreme maximum. Trends are significant with * $P < 0.10$, ** $P < 0.05$, *** $P < 0.01$. BBCH-code (see footnote 1): R_{00} : seeding, R_{09} : emergence, R_{31} : beginning of stem elongation, R_{51} : beginning of heading, R_{65} : full flowering, R_{89} : beginning of harvest (fully ripe), DOY: day of the year (BBCH: recommended scale for phenological observations according to [Strauß et al., 1994](#)).

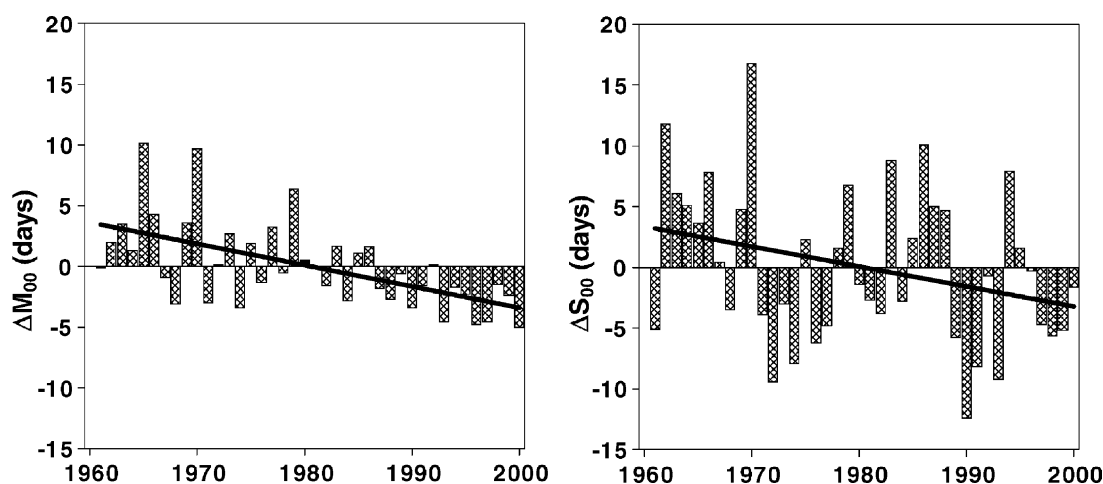


Fig. 7. Changes in the seeding dates of maize (left) and sugar beet (right) in the main cropping areas of Germany, 1961–2000 with calculated trends. Anomalies are shown.

shooting is always a good prerequisite for a high crop density at harvest. But until now the relatively moderate changes in phenology did not have any negative effects on the final crop density, because during the mild winters in the last decade, tillering was continued also in winter, so that the number of tillers was not limited. A longer period from shooting to heading is also advantageous for the very complex yield forming processes in this phase.

Main cropping areas for maize are the southern and western states of Germany. Here the temperatures are the best to grow this C_4 -crop. Between the individual states, differences in average timing of phenological events are low (Müller, 2002). For maize both phenophases in spring (M_{00} , M_{09}) are clearly advanced (Table 5). Most significant changes occurred since 1987 (Fig. 7). Since then higher temperatures in spring made an earlier sowing possible. The strong increasing temperatures in May (see Table 1) led additionally to an accelerated plant development, so that the date of emergence showed the strongest trend (–3.3 days per decade). As a result the period from seeding to emergence is clearly shortened. The beginning of harvest shows no significant trend, so that the time from emergence to harvest has extended by 2 days per decade.

With about 6 months, sugar beet has a relatively long growing time. Here also all early phases (S_{00} ,

S_{09} , S_{31}) show negative trends, however, of reduced significance (Table 6). The reason for the earlier seeding date is also the changed beginning of growing season. The advanced emergence and the earlier beginning of row closing seems to be only a result of earlier sowing dates. Because of the earlier row closing, the period until harvest is slightly extended, which is always favourable for the growth of the beet and the final yield.

Table 5

Statistical parameters of maize phenology in Germany (Federal States: Lower Saxony, North-Rhine Westphalia, Baden Württemberg, Bavaria), 1961–2000 and calculated trends

Phase (BBCH-code)	$\bar{x}$ (DOY)	Date	s (days)	Min (DOY)	Max (DOY)	Trend (days/10a)
M_{00}	119.9	30/4	4.5	115	130	–1.7***
M_{09}	136.0	16/5	5.6	124	154	–3.3***
M_{89}	269.6	27/9	8.8	259	284	–1.3
Duration of phases						
M_{00} – M_{09}	16.6		3.1	9	120	–1.6***
M_{09} – M_{89}	133.3		7.2	23	148	+2.1**

$\bar{x}$: mean, s : standard deviation, min: extreme minimum, max: extreme maximum. Trends are significant with * $P < 0.10$, ** $P < 0.05$, *** $P < 0.01$. BBCH-code (see footnote 1): M_{00} : seeding, M_{09} : emergence, M_{89} : beginning of harvest (fully ripe), DOY: day of the year (BBCH: recommended scale for phenological observations according to Strauß et al., 1994).

Table 6

Statistical parameters of sugar beet phenology in Germany (Federal States: Lower Saxony, Bavaria, North-Rhine Westphalia), 1961–2000 and calculated trends

Phase (BBCH- code)	x (DOY)	Date	s (days)	Min (DOY)	Max (DOY)	Trend (days/10a)
S ₀₀	99.0	9/4	6.9	87	116	−1.6*
S ₀₉	119.8	30/4	5.5	110	133	−1.2*
S ₃₁	167.8	17/6	5.2	156	177	−1.7*
S ₈₉	277.9	5/10	3.2	270	283	−0.6
Duration of phases						
S ₀₀ –S ₀₉	20.8		2.9	16	27	+0.4
S ₀₉ –S ₃₁	48.0		4.4	42	60	−0.5
S ₃₁ –S ₈₉	110.1		4.5	103	122	+1.2*
S ₀₉ –S ₈₉	158.1		5.2	148	167	+0.7

x: mean, s: standard deviation, min: extreme minimum, max: extreme maximum. Trends are significant with * $P < 0.10$, ** $P < 0.05$, *** $P < 0.01$. BBCH-code (see footnote 1): S₀₀: seeding, S₀₉: emergence, S₃₁: beginning of row closing, S₈₉: beginning of harvest (fully ripe), DOY: day of the year (BBCH: recommended scale for phenological observations according to Strauß et al., 1994).

4. Discussion

This investigation showed that not only the natural vegetation, in this study used to define the beginning of growing season, responded to the distinct climate changes in the 1990s, but also fruit trees and even annual crops like winter rye, maize and sugar beet. All plants showed an advanced timing of phenophases, mainly in spring, which was well correlated with the beginning of growing season. The mean air temperature from February to April was a good index which explained the annual variability of phenological phases in this time of the year. An earlier blossom of fruit trees holds the danger of damage by late frosts. Frosts before the beginning of blossom may cause masked injuries in flower buds, but the damage is less than it would be in the period of blossom. Frost during the flowering period can harm the blossoms, so that total crop failures can occur. The year 1981 was a good example for the strong impact of late frost damages on apple yields in Europe. Thus, it is very important to continue plant monitoring in the future, to be prepared for the impacts of climate change on agriculture and horticulture.

The currently observed changes in the timing of phenological events are still moderate and mostly

not more than a few days per decade, but the relationships between plant development and air temperature changes seems to be clear. The expected climate changes in this century will be probably much larger than those observed during the 20th century (Houghton et al., 2001). For Germany the increase in annual air temperature could range between 0.1 °C per decade (B1-scenario) and 0.45 °C per decade (A2-scenario, Hulme and Sheard, 1999). According to this scenario, the strongest seasonal changes in air temperature will occur in winter and in summer. For the spring development of plants mainly the temperature changes in winter and early spring are important, which steer the period of dormancy and ontogenetic development. The threat of late spring frosts, combined with more frequent mild winters, pose a challenge for even frost hardy species (Burroughs, 2002).

5. Conclusions

The results show that it is necessary to develop phenological models in order to estimate the impact of climate change on plant development in different regions of the world. For the natural vegetation already some progress was achieved. But this paper shows that all plants will react to climate changes in a similar way. Therefore, in the future more efforts are necessary to estimate the impacts of temperature rise on fruit trees, horticultural plants and agricultural crops. These estimations are possible and much easier to derive than reliable estimates of future crop yield changes.

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